

# TIM YANG

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Siebel School of Computing and Data Science  
2318 Siebel Center  
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## EDUCATION

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### University of Illinois Urbana-Champaign (UIUC)

|                  |                                                         |       |
|------------------|---------------------------------------------------------|-------|
| PhD, Kinesiology | Computational Rehabilitation for Power Wheelchair Users | ≈2025 |
| Certificate      | Information Accessibility, Design, and Policy           | 2019  |
| Certificate      | Foundations in Teaching                                 | 2017  |

### University of Central Oklahoma

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|----------------------|------|
| BS, Computer Science | 2012 |
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## POSITIONS

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### School of Computing and Data Science, UIUC

|                                |           |
|--------------------------------|-----------|
| Senior Instructional Developer | 2023+     |
| Instructional Developer        | 2020–2023 |

### Microsoft Lighthouse Program, UIUC

|                 |           |
|-----------------|-----------|
| Doctoral Fellow | 2018–2020 |
|-----------------|-----------|

### Department of Health and Kinesiology, UIUC

|                    |           |
|--------------------|-----------|
| Research Assistant | 2012–2018 |
| Teaching Assistant | 2015–2017 |

## HONORS

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### Fellowships

|                      |                                              |           |
|----------------------|----------------------------------------------|-----------|
| Doctoral Fellowship  | Microsoft Lighthouse Program, UIUC           | 2018–2020 |
| Research Scholarship | NIH INBRE Program, OU Health Sciences Center | Su 2011   |

### Awards

|                       |                                                         |      |
|-----------------------|---------------------------------------------------------|------|
| Finalist              | Image of Research Competition, UIUC                     | 2019 |
| 1 <sup>st</sup> Place | Research Live Competition, UIUC                         | 2018 |
| Best Paper            | Student Paper Competition, RESNA Conference             | 2015 |
| Honorable Mention     | Student Paper Competition, RESNA Conference             | 2015 |
| 2 <sup>nd</sup> Place | Computational Science and Engineering Competition, UIUC | 2014 |
| Honorable Mention     | Student Paper Competition, RESNA Conference             | 2014 |
| Outstanding Research  | Department of Computer Science, UCO                     | 2012 |

## RESEARCH

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### University of Illinois

Rehabilitation Engineering Laboratory

2012–2018

- Pressure ulcer risk Predicted pressure ulcer risk from seat pressure via AI/ML
- Power mobility risk Assessed power wheelchair driving via probabilistic AI/ML
- Personalized seating Assessed inflation patterns via custom programmable cushion
- Joystick control Extracted muscle synergies for power wheelchair driving via AI/ML
- Cloud monitoring Tracked power mobility usage via custom cloud dashboard
- Blood flow response Assessed reactive hyperemia in spinal injury via laser Doppler

### University of Oklahoma Health Sciences Center

Rehabilitation Biomechanics Laboratory

2011–2012

- Wheelchair tilt/recline Assessed seat pressure vs tilt/recline via ANOVA
- Wheelchair tracking Assessed free-living wheelchair usage via accelerometry

NIH Summer Research Program

Su 2011

- Soft tissue indentation Assessed blood flow response vs mechanical indentation
- Ulcer-cytokine model Tested cytokine response vs ulceration in rats via ELISA

## PUBLICATIONS

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- 7 Ren S, Chen Z, Qin X, Zhao X, Yang TD, Zhu W. Measurement and evaluation of bone loading in physical activity: a systematic review. *Meas Phys Educ Exerc Sci*. 2021;25(2):149–162
- 6 Yang TD, Jan YK. Predicting pressure ulcer risk from seating interface pressure using nonnegative matrix factorization. *Med Biol Eng Comput*. 2020;58:227–237
- 5 Lung CW, Yang TD, Liao BY, Cheung WC, Jain S, Jan YK. Dynamic changes in seating pressure gradients in people with spinal cord injury. *Assist Technol*. 2020;32(5):277–286
- 4 Liao F, Yang TD, Wu FL, Cao CM, Mohamed A, Jan YK. Using multiscale entropy to assess the efficacy of local cooling on reactive hyperemia in people with spinal cord injury. *Entropy*. 2019;21(1):90 (12 pages)
- 3 Chen Y, Wang J, Lung CW, Yang TD, Crane B, Jan YK. Effect of wheelchair tilt and recline on ischial and coccygeal interface pressure in people with spinal cord injury. *Am J Phys Med Rehabil*. 2015;93(12):1019–1030
- 2 Lung CW, Yang TD, Crane B, Elliott J, Dicianno BE, Jan YK. Investigation of peak pressure index parameters for people with spinal cord injury using wheelchair tilt and recline: methodology and preliminary report. *Biomed Res Int*. 2014;2014:508583 (9 pages)
- 1 Yang TD, Hutchinson S, Rice LA, Watkin KL, Jan YK. Development of a scalable wireless monitoring system for wheelchair tilt usage. *Int J Phys Med Rehabil*. 2013;1(4):129 (6 pages)

## PRESENTATIONS

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### Talks

- 6 Yang TD. On the road to self-driving ... drivers? Presented at: *Research Live Competition*; October 2018; Urbana, IL
  - Award: 1<sup>st</sup> Place

- 5 Yang TD. Decoding pressure ulcer risk from seating interface pressure using matrix factorization. Presented at: *Rehabilitation Science Seminar*; February 2018; UIUC, Champaign, IL
- 4 Yang TD, Rice LA, David A, Hutchinson S, Jan YK. Myoelectric modeling of joystick control for adaptive smart wheelchairs. Presented at: *36<sup>th</sup> RESNA Conference*; June 2015; Denver, CO
  - Award: Best Paper
- 3 Yang TD, Patil A, Jan YK. Individualized performance quantification of power wheelchair driving. Presented at: *35<sup>th</sup> RESNA Conference*; June 2014; Indianapolis, IN
  - Award: Honorable Mention Paper
- 2 Yang TD. Markov modeling of power wheelchair driving. Presented at: *Rehabilitation Science Seminar*; October 2013; UIUC, Champaign, IL
- 1 Yang TD, Liao F, Jones MA, Jan YK. Effect of wheelchair tilt and recline on peak seating pressure in people with spinal cord injury. Presented at: *28<sup>th</sup> Southern Biomedical Engineering Conference*; May 2012; University of Texas MD Anderson Cancer Center, Houston, TX

## Posters

- 10 Yang TD, Rice LA, Hutchinson S, Jan YK. Robotic Individualized Driving Evaluation (RIDE): design and preliminary evaluation. Presented at: *19<sup>th</sup> IEEE International Conference on Rehabilitation Robotics*; May 2025; Chicago, IL
- 9 Yang TD, Rice LA, Jan YK. Typifying power wheelchair joystick control using EMG feature engineering and visualization. Presented at: *39<sup>th</sup> RESNA Conference*; July 2018; Arlington, VA
- 8 Jan YK, Lung CW, Yang TD, Cheung W, Jain S. Seating pressure gradient vectors in response to wheelchair tilt and recline in people with spinal cord injury. Presented at: *93<sup>rd</sup> American Congress on Rehabilitation Medicine*; November 2016; Chicago, IL
- 7 Yang TD, Kibler K, Lung CW, Jan YK. Development and evaluation of a programmable alternating pressure seat cushion. Presented at: *36<sup>th</sup> RESNA Conference*; June 2015; Denver, CO
  - Award: Honorable Mention Paper
- 6 Yang TD, Hutchinson S, Jan YK. Markov framework for power wheelchair driving. Presented at: *3<sup>rd</sup> Computational Science and Engineering Meeting*; April 2014; UIUC, Urbana, IL
  - Award: 2<sup>nd</sup> Place
- 5 Yang TD, Hutchinson S, Rice LA, Watkin KL, Jan YK. Pressure ulcer prevention via Raspberry Pi. Presented at: *Center on Health, Aging, and Disability Symposium*; March 2013; UIUC, Champaign, IL
- 4 Yang TD, Liao F, Jones MA, Jan YK. Sitting-induced pressure ulcer risks may be reduced at specific tilt and recline angles. Presented at: *NIH INBRE Symposium*; July 2012; OUHSC, Oklahoma City, OK
- 3 Yang TD, Liao F, Jones MA, Jan YK. Effect of wheelchair tilt and recline on peak seating pressure in people with spinal cord injury. Presented at: *Allied Health Research Day*; April 2012; OUHSC, Oklahoma City, OK
- 2 Yang TD, Liao F, Jones MA, Jan YK. Effect of wheelchair tilt and recline on peak seating pressure in people with spinal cord injury. Presented at: *Graduate Research Education and Technology Symposium*; April 2012; OUHSC, Oklahoma City, OK
- 1 Yang TD, Fu J, Jones MA, Jan YK. Quantifying free-living power wheelchair usage using accelerometry. Presented at: *Oklahoma Research Day*; November 2011; Cameron University, Lawton, OK

## TEACHING

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### Department of Health and Kinesiology, UIUC

|                    |                                             |           |
|--------------------|---------------------------------------------|-----------|
| Guest Lecturer     | Graduate Academy for College Teaching       | 2020      |
|                    | Physical Activity Research Methods (KIN201) | 2018      |
|                    | Teaching Professionals Program (TPR02)      | 2018      |
| Lab Instructor     | Rehabilitation Biomechanics (KIN494)        | 2015–2017 |
| Teaching Assistant | Drug Use and Abuse (CHLH243)                | 2015      |

### International Graduate Mentor Program, UIUC

|        |                                          |           |
|--------|------------------------------------------|-----------|
| Mentor | 6 grad students (Rehabilitation Science) | 2018–2019 |
|--------|------------------------------------------|-----------|

### Independent Study, UIUC

|        |                                              |      |
|--------|----------------------------------------------|------|
| Mentor | 2 undergrad students (Kinesiology)           | 2018 |
| Mentor | 1 undergrad student (Mechanical Engineering) | 2014 |

### MoST Scholars Program, UIUC

|        |                                               |           |
|--------|-----------------------------------------------|-----------|
| Mentor | 6 undergrad students (Biomedical Engineering) | 2014–2016 |
|--------|-----------------------------------------------|-----------|

### Khorana Scholars Program, UIUC

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|--------|----------------------------------------------|---------|
| Mentor | 1 grad student (Bioengineering)              | Su 2014 |
| Mentor | 1 undergrad student (Electrical Engineering) | Su 2013 |

## GRANTS

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### Completed

- 2 13288 UIUC Campus Research Board. Wheelchair tilt and recline for preventing pressure ulcers in people with spinal cord injury. Cost: \$25,000. PI: Jan YK; 2013–2014
  - Role: Research Assistant (first/co-authored 2 peer-reviewed publications)
- 1 R03HD060751 NIH Eunice Kennedy Shriver National Institute of Child Health and Human Development. Effect of power seating on tissue viability in wheelchair users with spinal cord injury. Cost: \$165,500. PI: Jan YK; 2011–2012
  - Role: Research Assistant (co-authored 2 peer-reviewed publications)

### Unfunded

- 2 D2 NIDILRR Rehabilitation Engineering Research Center. Development of a pervasive power seating framework. PI: Hutchinson SA, Jan YK; 2013
  - Role: Research Assistant (drafted complete proposal)
- 1 D1 NIDILRR Rehabilitation Engineering Research Center. Development of a pervasive power mobility framework. PI: Hutchinson SA, Jan YK; 2013
  - Role: Research Assistant (drafted complete proposal)

## SERVICE

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### Intramural

|                       |                                              |       |
|-----------------------|----------------------------------------------|-------|
| Accessibility Liaison | Illinois Accessibility Liaison Program, UIUC | 2019+ |
|-----------------------|----------------------------------------------|-------|

|                     |                                                             |                  |
|---------------------|-------------------------------------------------------------|------------------|
| Judging Committee   | Research Live Competition, UIUC                             | 2018             |
| English Coach       | Gies College of Business, UIUC                              | 2018             |
| Grader              | Illinois Math Finals, Department of Mathematics, UIUC       | 2014–2016        |
| <b>Extramural</b>   |                                                             |                  |
| Reviewer            | ICORR Conference                                            | 2024+            |
| Reviewer            | PLOS ONE                                                    | 2018+            |
| Reviewer            | RESNA Conference                                            | 2015–2019        |
| <b>Professional</b> |                                                             |                  |
| Member              | IEEE (Institute of Electrical and Electronics Engineers)    | 2013–2015, 2024+ |
| Member              | EMBS (IEEE Engineering in Medicine and Biology Society)     | 2013–2015, 2024+ |
| Member              | ICORR (International Consortium of Rehabilitation Robotics) | 2024+            |
| Member              | RESNA (Rehabilitation Engineering Society of North America) | 2013–2019        |